

Population Agglomeration and Spatial Reallocation in China

Li Shen*

Working Paper — Comments Welcome
April 2026

Abstract

Between 2002 and 2020, China added roughly 140 million people while undergoing one of the fastest urbanization episodes in recorded history, but few studies document the full national population distribution at fine spatial resolution over this period. I use annual 5 km population density rasters covering mainland China to compute concentration measures, a population-weighted centroid trajectory, and provincial convergence diagnostics for each of 19 years. Several indicators point to increased spatial agglomeration: the spatial Gini rose from 0.852 to 0.868, and the share of total population in the densest decile of pixels increased from 73.5 to 76.7 percent. The geographic footprint of high-density areas expanded by 24 percent, with gains sharply clustered around the Yangtze River Delta, Pearl River Delta, and Beijing–Tianjin corridor. The population-weighted centroid drifted 17 km to the southeast. I find no evidence of systematic provincial catch-up: the β -convergence coefficient is positive and insignificant, while cross-provincial dispersion widened. The evidence is more consistent with agglomeration and selective core expansion than with province-level equalization.

Keywords: population geography, agglomeration, spatial economics, regional convergence, China

JEL Codes: R12, R23, O18, J11

1 Introduction

Between 2002 and 2020, China added roughly 140 million people to a population that was already the world’s largest, while undergoing rapid urbanization and changes in transport and

*Department of Economics, Clark University. Email: LShen@clarku.edu. All errors are my own.

market integration.¹ A large and growing literature in spatial economics studies the forces that shape such transformations—market access, agglomeration economies, transport costs, and migration frictions (Redding and Rossi-Hansberg, 2017)—and has developed quantitative frameworks capable of linking these forces to the observed distribution of population and production across space.² One empirical question has received less systematic attention: at a fine spatial resolution, how did the distribution of China’s population change during this period?

The answer is not obvious from existing evidence. Province-level data show that China’s coastal regions grew faster than the interior, but provinces are large and heterogeneous: a province can appear stable on average while its population becomes sharply concentrated in a few urban corridors and metropolitan fringes. City-level studies illuminate individual metropolitan areas, but they cannot characterize the national geography of gains and losses across the full territory.³ The quantitative spatial models that could deliver such an account require structural estimation and are typically calibrated to a single cross-section rather than used to describe long-run trajectories.⁴ To date, the joint evolution of population concentration, core expansion, directional movement, and provincial catch-up has received limited systematic documentation at this spatial scale.

This paper offers such an account. I ask: what are the main spatial facts of China’s population reallocation between 2002 and 2020? Rather than identifying the causal effect of a particular policy or estimating a structural model, I construct a set of spatial indicators that characterize the national geography of population, and I interpret those indicators using the framework of spatial economics.

The empirical strategy combines high-resolution raster data with standard spatial-economics diagnostics. I use annual population density rasters from WorldPop (WorldPop and CIESIN, 2018), which disaggregate census counts to a fine grid using a random-forest model trained on satellite imagery, land cover, and road networks (Stevens et al., 2015). The source rasters

¹China’s urbanization rate rose from about 39 percent in 2002 to 64 percent in 2020 (National Bureau of Statistics). On transport expansion, see Zheng and Kahn (2013). On the extent of domestic market fragmentation and its evolution, see Poncet (2005). For a comprehensive account of the structural forces behind this transformation, see Brandt et al. (2014).

²Canonical contributions include Krugman (1991), Fujita et al. (1999), Allen and Arkolakis (2014), and Ahlfeldt et al. (2015). Caliendo et al. (2018) extend the framework to multiple sectors and regions. See Redding and Rossi-Hansberg (2017) for a survey.

³Au and Henderson (2006) study city-size efficiency; Baum-Snow et al. (2017) examine within-city decentralization driven by transport infrastructure; Chauvin et al. (2017) compare agglomeration patterns across China, India, Brazil, and the United States. These are important contributions, but few provide a consistent national account of how the full population-density surface evolved year by year.

⁴For example, Tombe and Zhu (2019) calibrate a multi-region trade-and-migration model to a single year; Desmet and Rossi-Hansberg (2014) study spatial dynamics but in a stylized geography. The present paper complements these contributions by documenting the spatial facts that such models aim to explain.

are available at approximately 1 km resolution; I downsample them to a 5 km analytical grid and restrict the sample to mainland China using a province-boundary mask that excludes Hong Kong, Macao, and Taiwan. On this grid I compute pixel-level concentration measures (a spatial Gini coefficient, top-decile population shares, high-density pixel counts), a population-weighted centroid trajectory, and province-level convergence diagnostics (β - and σ -convergence) following [Barro and Sala-i Martin \(1992\)](#). All indicators are computed annually from 2002 to 2020, yielding 19 cross-sections.

I report four findings. First, spatial agglomeration intensified. The spatial Gini coefficient—a measure of how unevenly population is distributed across pixels—rose from 0.852 in 2002 to 0.868 in 2020 (Table 1, Figure 1). The share of total population living in the densest 10 percent of pixels increased from 73.5 to 76.7 percent over the same period. Both indicators trend upward over the full sample, suggesting that population growth was absorbed disproportionately by already dense locations rather than spread evenly across space.

Second, the geographic footprint of high-density areas expanded. The number of pixels exceeding 1,000 persons per km² grew from 10,391 to 12,894—an increase of 24 percent—while the population-weighted mean density rose from 140 to 155 persons per km² (Figure 2). A cumulative change map (Figure 3) shows that density gains were sharply clustered around the Yangtze River Delta, Pearl River Delta, Beijing–Tianjin corridor, and Chengdu–Chongqing axis, while much of the interior and northeast changed little or lost density. Of the roughly 449,000 inhabited mainland pixels, about 240,000 gained density and 208,000 lost it, but density gains were larger in magnitude, on average, than density losses.

Third, the population-weighted centroid of mainland China shifted approximately 17 km toward the southeast, from (32.36°N, 113.71°E) to (32.24°N, 113.82°E), with a southward component of 13 km and an eastward component of 11 km (Figure 4). Together with the cumulative change map, the centroid drift is consistent with relative population gains in coastal and south-central provinces.

Fourth, there is no evidence of provincial convergence. The cross-sectional regression of annualized density growth on log initial density yields a slope of $\hat{\beta} = 0.10$ ($p = 0.41$, $R^2 = 0.02$; Figure 5, Table 2), meaning that initially sparse provinces did not grow faster than initially dense ones. The coefficient of variation of provincial mean densities increased monotonically from 1.18 to 1.54 over the sample period, providing no evidence of σ -convergence. The cross-provincial density distribution fanned out rather than compressed.

This paper contributes to three literatures. The first is the quantitative spatial economics literature that studies how agglomeration, market access, and trade costs shape the distribution of population and economic activity across space. The theoretical core of this literature—running from [Krugman \(1991\)](#) through [Allen and Arkolakis \(2014\)](#), [Ahlfeldt et al.](#)

(2015), and [Caliendo et al. \(2018\)](#)—generates predictions about spatial concentration, core-periphery dynamics, and the role of transport infrastructure. Empirical work has tested these predictions using natural experiments such as German reunification ([Redding and Sturm, 2008](#)), the Tennessee Valley Authority ([Kline and Moretti, 2014](#)), and American railroads ([Donaldson and Hornbeck, 2016](#)). I do not estimate a structural model or exploit a natural experiment. I complement this literature by providing a descriptive benchmark against which such models can be evaluated: how concentration, core expansion, and centroid drift evolved in one of the world’s largest spatial economies over two decades.

The second is the literature on China’s internal economic geography, including work on market fragmentation ([Poncet, 2005](#)), city-size efficiency ([Au and Henderson, 2006](#)), transport infrastructure and spatial structure ([Baum-Snow et al., 2017](#); [Faber, 2014](#); [Zheng and Kahn, 2013](#)), and the quantification of goods- and labor-market frictions ([Tombe and Zhu, 2019](#)).⁵ A related strand studies internal migration under institutional constraints, including the hukou system ([Chan and Zhang, 1999](#)), and its interaction with firm-level labor demand ([Imbert et al., 2022](#)).⁶ I document the resulting spatial distribution of population—the pattern that these frictions and policies jointly produced—at a resolution finer than the administrative units used in this literature.

The third is the use of remotely sensed and model-based spatial data in economics. [Henderson et al. \(2012\)](#) show that satellite nighttime lights can proxy for economic growth; [Lessmann and Seidel \(2017\)](#) use nightlights to study regional convergence across countries; and [Dingel et al. \(2021\)](#) construct metropolitan-area boundaries for developing economies from light-based contiguity. The WorldPop rasters I use ([Stevens et al., 2015](#)) are a parallel innovation on the population side. I apply this measurement approach to a single country over time, producing a nationally consistent set of spatial facts rather than a cross-country comparison.⁷

Standard convergence tests applied to provincial aggregates can miss the spatial restructuring documented here: the national density distribution became more concentrated and more dispersed across provinces at the same time.

The paper proceeds as follows. Section 2 describes the data. Section 3 defines the

⁵[Faber \(2014\)](#) shows that highway integration can concentrate rather than diffuse industrial activity, a mechanism consistent with the rising concentration I document. [Hsieh and Moretti \(2019\)](#) demonstrate that housing-supply constraints generate spatial misallocation of labor in the United States; an analogous mechanism—operating through the hukou system rather than zoning—may be relevant in China.

⁶[Bryan and Morten \(2019\)](#) estimate aggregate productivity effects of internal migration barriers in Indonesia. [Kennan and Walker \(2011\)](#) develop a structural model of individual migration decisions driven by expected income differences.

⁷[Michaels et al. \(2012\)](#) link urbanization to structural transformation using cross-country data; my analysis operates within a single country but at finer spatial resolution.

empirical framework. Section 4 presents the results. Section 5 discusses the results through a spatial-economics lens, and Section 6 concludes.

2 Data

The core data source is the WorldPop UN-adjusted annual population density raster product for China (WorldPop and CIESIN, 2018). WorldPop constructs these layers by disaggregating census and administrative population counts to a fine grid using a random-forest model that incorporates satellite imagery, land cover, road networks, and other geospatial covariates (Stevens et al., 2015). The source rasters are available at approximately 1 km resolution (30 arc-seconds) and cover the period from 2000 to 2020. I use the 2002–2020 subset, yielding 19 annual layers.

For computational stability and for integration into a web-based evidence explorer, I downsample the raw 1 km rasters to an analytical grid of approximately 5 km (0.0417 degrees per pixel). Downsampling is performed by averaging density values within each output cell. The resulting grid contains 547,888 valid pixels covering mainland China and preserves national coverage and annual comparability while remaining light enough for repeated statistical summaries and interactive visualization. This resolution is adequate for national- and province-level patterns, though it is not designed for neighborhood-level claims.

The analysis is organized at two levels. At the pixel level, I study concentration, cumulative change, and centroid movement. At the province level, I aggregate raster values using a publicly available province-boundary GeoJSON keyed by six-digit administrative codes (`adcode`).⁸ This yields mean density, estimated population, and growth rates for each of 31 mainland provinces. Many spatial processes become visible at the grid-cell level but are discussed in regional-development terms at the province level, so both scales are needed.

Two geographic scopes are maintained throughout the workflow. The first includes Hong Kong, Macao, and Taiwan. The second restricts attention to mainland China (31 provinces) and serves as the main analytical sample. The mainland scope aligns with conventional regional-development discussions, and all results reported in the paper use this scope unless noted otherwise.

The data should be interpreted as a spatial measurement framework rather than as exact census counts. Raster population products are model-based and therefore contain uncertainty (Stevens et al., 2015). Their main value lies in the ability to describe broad spatial

⁸The boundary file contains 34 provincial-level features with standard administrative codes (e.g., 110000 for Beijing). It is widely used in Chinese spatial analysis and is consistent with NBS province delineations, but it is not an official NBS release.

patterns consistently over time. They are especially useful for measuring concentration and directional change at the national scale, although they are less well suited to strong statements about micro-level mechanisms. Table 1 reports key summary statistics for selected years.

3 Empirical Framework

I construct three groups of indicators: pixel-level concentration, directional reallocation, and cross-provincial heterogeneity.

3.1 Pixel area and population estimation

Each pixel in the analytical grid corresponds to a square of approximately $\Delta = 0.0417$ degrees (roughly 5 km at mid-latitudes). Because the metric area of a degree of longitude varies with latitude, I correct pixel area using a cosine adjustment:

$$a_i = (\Delta \times 111.32)^2 \cos(\phi_i),$$

where ϕ_i is the latitude of pixel i in radians and 111.32 is the approximate number of kilometers per degree of latitude. Pixel-level population is then estimated as

$$p_{it} = d_{it} a_i,$$

where d_{it} denotes population density (persons per km²) in pixel i and year t . Across mainland China, the grid contains 547,888 valid pixels, of which approximately 448,600 are inhabited (density > 0) in any given year.

Pixels with missing or negative values (arising from ocean masking or data gaps in the source rasters) are set to NaN and excluded from all computations. This nodata treatment is applied consistently across all years.

3.2 Spatial concentration

Using the distribution of p_{it} across valid pixels, I compute two complementary concentration measures. The first is the spatial Gini coefficient. Let $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(n)}$ denote the sorted pixel populations in year t , and let $S_k = \sum_{j=1}^k p_{(j)}$ be the cumulative sum up to rank

k . The Gini coefficient is then

$$G_t = 1 - \frac{2 \sum_{k=1}^n S_k}{n \cdot S_n} + \frac{1}{n}.$$

Values near 1 indicate extreme concentration; values near 0 indicate spatial uniformity.

The second measure is the top-decile concentration share: the fraction of total national population residing in the densest 10 percent of pixels:

$$C_t^{10} = \frac{\sum_{k=1}^{\lfloor 0.1n \rfloor} p_{(n-k+1)}}{\sum_{k=1}^n p_{(k)}}.$$

I also track the number of pixels exceeding a fixed density threshold of 1,000 persons per km^2 as a simple measure of core-area expansion over time.

3.3 Directional reallocation

For each year, I compute the population-weighted centroid of the national population distribution:

$$\bar{x}_t = \frac{\sum_i p_{it} x_i}{\sum_i p_{it}}, \quad \bar{y}_t = \frac{\sum_i p_{it} y_i}{\sum_i p_{it}},$$

where (x_i, y_i) denotes the longitude and latitude of pixel i . Changes in the centroid over time summarize the macro direction of spatial population movement. The total displacement is computed using a Euclidean approximation with the same latitude-dependent scaling used for pixel area.

3.4 Cross-provincial convergence

For each province p , I define the provincial mean density as the simple average of pixel-level densities across the n_p pixels that fall within the province boundary:

$$\bar{d}_{pt} = \frac{1}{n_p} \sum_{i \in p} d_{it}.$$

Because the analytical grid is on a regular latitude–longitude mesh, pixels at different latitudes represent slightly different metric areas. The simple pixel average is therefore not identical to the area-weighted mean $\hat{d}_{pt} = \sum_{i \in p} p_{it} / \sum_{i \in p} a_i$, which would be the standard ratio of total population to total land area. For provinces spanning a narrow latitude range, the difference is negligible (typically <1%). For high-latitude provinces with large north-south extent, the discrepancy can reach a few percent, but this does not affect the convergence re-

sults: re-estimating the β -convergence regression with area-weighted means produces a slope of 0.102 ($p = 0.416$), essentially identical to the baseline (replication code computes both specifications).⁹ I use the pixel average for consistency with the companion web application. I then use $\bar{d}_{pt}$ to construct convergence diagnostics following the framework of Barro and Sala-i Martin (1992).

β -convergence is assessed through the cross-sectional relationship

$$g_p = \alpha + \beta \ln \bar{d}_{p,2002} + \varepsilon_p,$$

where $g_p = (\bar{d}_{p,2020}/\bar{d}_{p,2002})^{1/18} - 1$ is the annualized growth rate of mean density over 18 years. A negative $\hat{\beta}$ would indicate that initially lower-density provinces experienced faster density growth, consistent with catch-up.

σ -convergence is summarized using the coefficient of variation of provincial mean densities:

$$CV_t = \frac{\text{sd}(\bar{d}_{pt})}{\text{mean}(\bar{d}_{pt})}.$$

A decline in CV_t over time would indicate a broad reduction in cross-provincial dispersion. Throughout the convergence analysis, Hong Kong, Macao, and Taiwan are excluded to maintain consistency with mainland administrative boundaries.

4 Results

4.1 Spatial concentration

The first and most robust pattern is rising spatial concentration. Over the 2002–2020 sample period, the spatial Gini coefficient increased from 0.852 to 0.868, a rise of roughly 1.8 percent (Figure 1, left axis). The top-decile population share—the fraction of the national population residing in the densest 10 percent of pixels—rose from 73.5 to 76.7 percent over the same period (Figure 1, right axis). Both indicators trend upward over the full sample, with minor year-to-year fluctuations, indicating that population growth was absorbed disproportionately by already dense locations.

The count of high-density pixels (exceeding 1,000 persons per km²) grew from 10,391 in 2002 to 12,894 in 2020—an increase of 24 percent—indicating that dense areas expanded in spatial extent as well as in intensity (Figure 2). The population-weighted mean density rose from 140.1 to 154.6 persons per km², so the average person lived in an increasingly dense

⁹The area-weighted R^2 is 0.023 versus 0.023 for the pixel-average specification. Both estimates are statistically insignificant and economically negligible in difference.

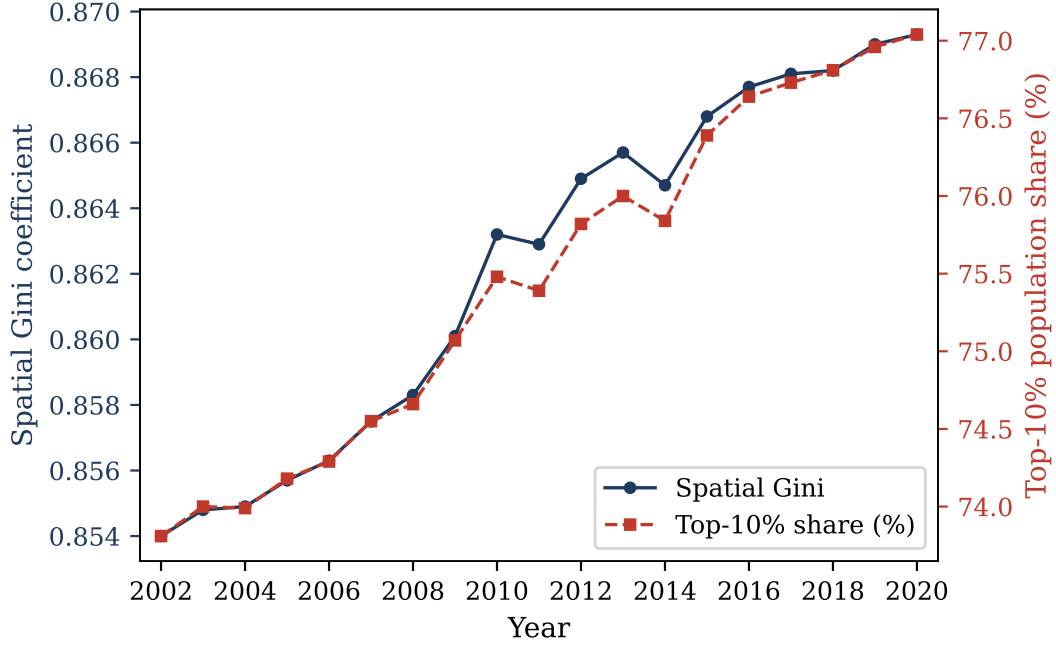


Figure 1: Spatial Gini coefficient and top-10% population concentration share, mainland China, 2002–2020.

environment.

Table 1 provides a broader statistical summary for selected years, confirming that the mainland estimated population grew from approximately 1.317 billion in 2002 to 1.453 billion in 2020 while all concentration measures increased.

Table 1: National summary statistics, mainland China (selected years)

Year	Est. pop. (100 million)	Mean density (pers./km ²)	Wtd. mean density	High-density pixels (>1000)	Spatial Gini	Top-10% share (%)
2002	13.17	133.7	140.1	10,391	0.8523	73.5
2005	13.40	136.1	142.6	10,831	0.8540	73.8
2010	13.80	140.1	146.9	11,852	0.8616	75.2
2015	14.19	144.1	151.0	12,448	0.8652	76.1
2020	14.53	147.4	154.6	12,894	0.8677	76.7

Notes: Population is estimated as $\sum_i d_{it} a_i$ in units of 10,000 persons, then converted to 100 million. Mean density is the unweighted average across valid pixels. Weighted mean density uses population-weighted averaging. High-density pixels count cells exceeding 1,000 persons/km². Spatial Gini and top-10% concentration are computed from the pixel-level population distribution. Data source: WorldPop UN-adjusted 1km rasters, downsampled to approximately 5km.

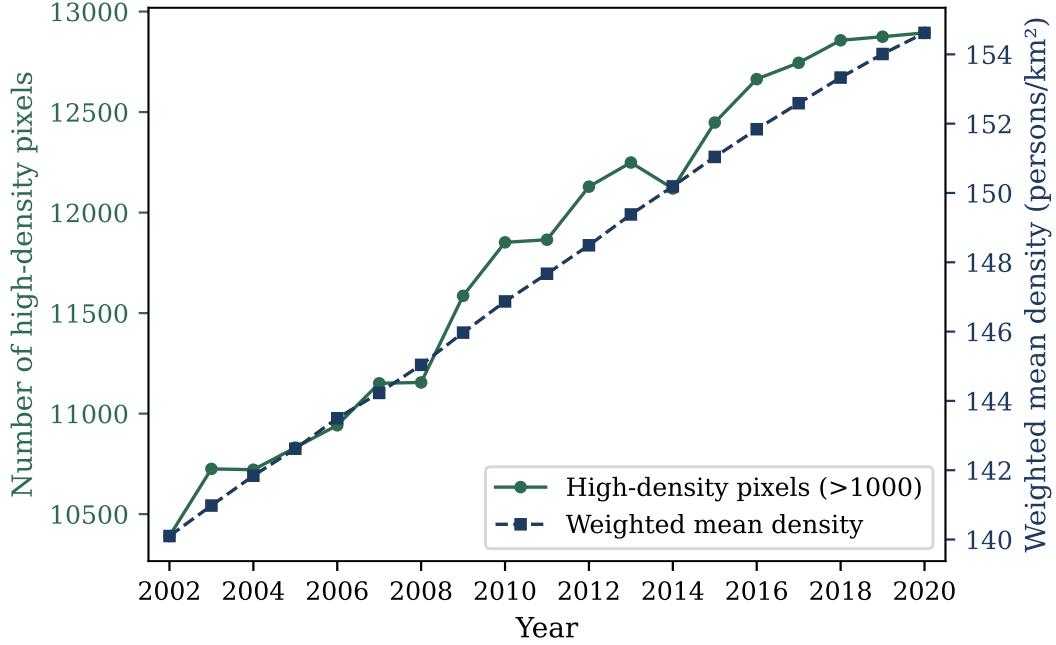


Figure 2: Number of high-density pixels ($>1,000$ persons/km²) and weighted mean density, mainland China, 2002–2020.

4.2 Cumulative change

Figure 3 maps the pixel-level difference in population density between 2020 and 2002. The national pattern is not one of smooth, broadly shared densification. Instead, large gains (red) are concentrated around already dense zones—the Yangtze River Delta, Pearl River Delta, Beijing–Tianjin corridor, and the Chengdu–Chongqing axis—while extensive areas of the interior and northeast changed little or lost density (blue). Of the approximately 449,000 inhabited mainland pixels, roughly 240,000 experienced density increases and 208,000 experienced decreases, but the gains were far larger in magnitude: mean density rose by 16.8 persons per km² across inhabited pixels.

This pattern is expected if core regions widen outward rather than if the national space converges toward a common density profile. In core-periphery models (Krugman, 1991; Fujita et al., 1999), dense, market-connected areas attract further inflows while peripheral areas lose relative standing. The geographic footprint of China’s dense cores expanded by roughly 2,500 high-density pixels over the sample period.

4.3 Centroid trajectory

The population-weighted centroid of mainland China moved from (32.36°N, 113.71°E) in 2002 to (32.24°N, 113.82°E) in 2020, a displacement of approximately 17 km toward the

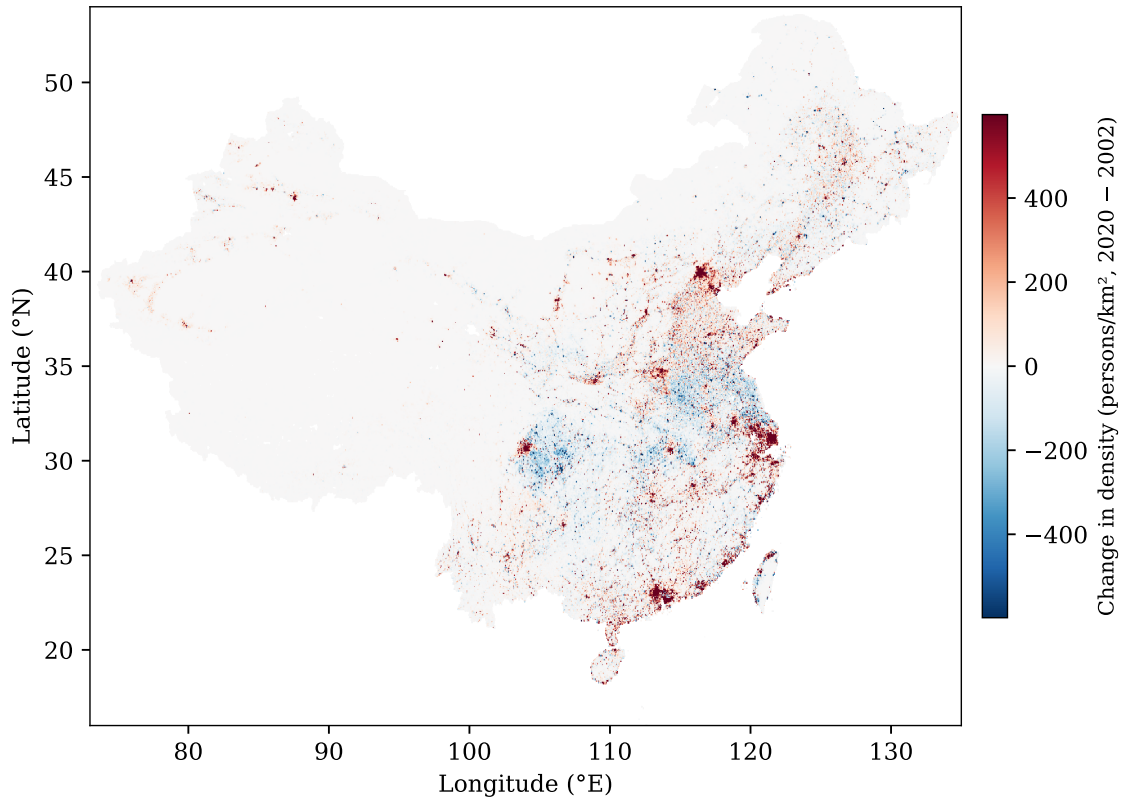


Figure 3: Cumulative change in population density, 2020 minus 2002 (persons/km²). Red indicates density gains; blue indicates losses. Color scale clipped at the 99th percentile of absolute change.

southeast (Figure 4). The latitudinal component of the shift was approximately 13 km southward, and the longitudinal component was approximately 11 km eastward.

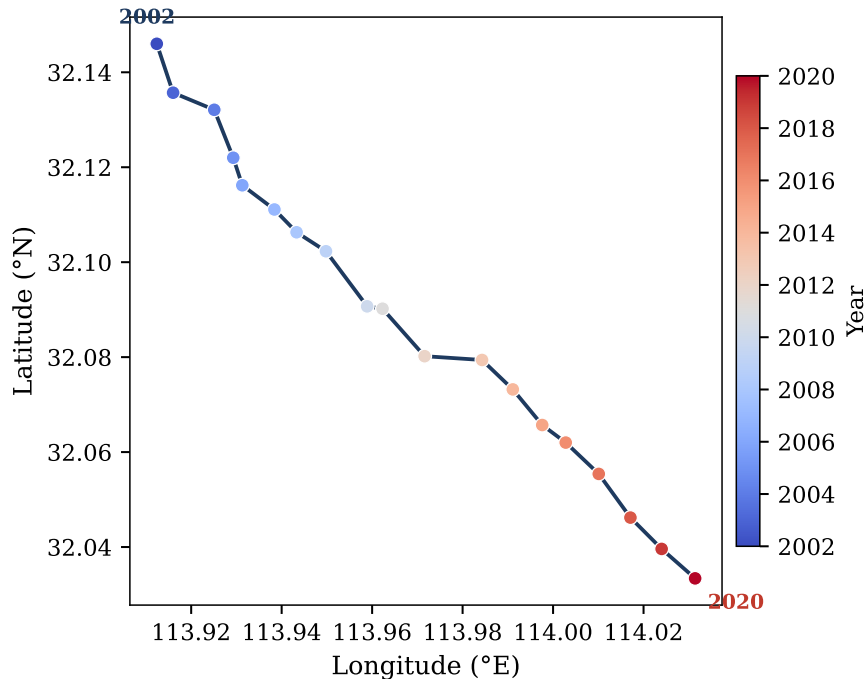


Figure 4: Population-weighted centroid trajectory, mainland China, 2002–2020. Each point represents one year, colored from blue (2002) to red (2020).

The southeastern direction reflects continued population growth in the coastal and south-central provinces—particularly Guangdong, Zhejiang, and Fujian—relative to the northern and western interior. The centroid compresses the full spatial distribution into a single trajectory that links pixel-level change to the macro geography of population.

4.4 Provincial convergence

At the province level, the evidence suggests no meaningful equalization during the sample period.

β -convergence. The cross-sectional relationship between log initial density and annualized growth rate yields an estimated slope of $\hat{\beta} = 0.102$ with a p -value of 0.415 and $R^2 = 0.023$ (Figure 5, Table 2). This coefficient is positive and statistically insignificant, meaning that there is no detectable tendency for initially lower-density provinces to grow faster than higher-density provinces. If anything, the point estimate suggests mild divergence, although the relationship is too weak to interpret strongly.

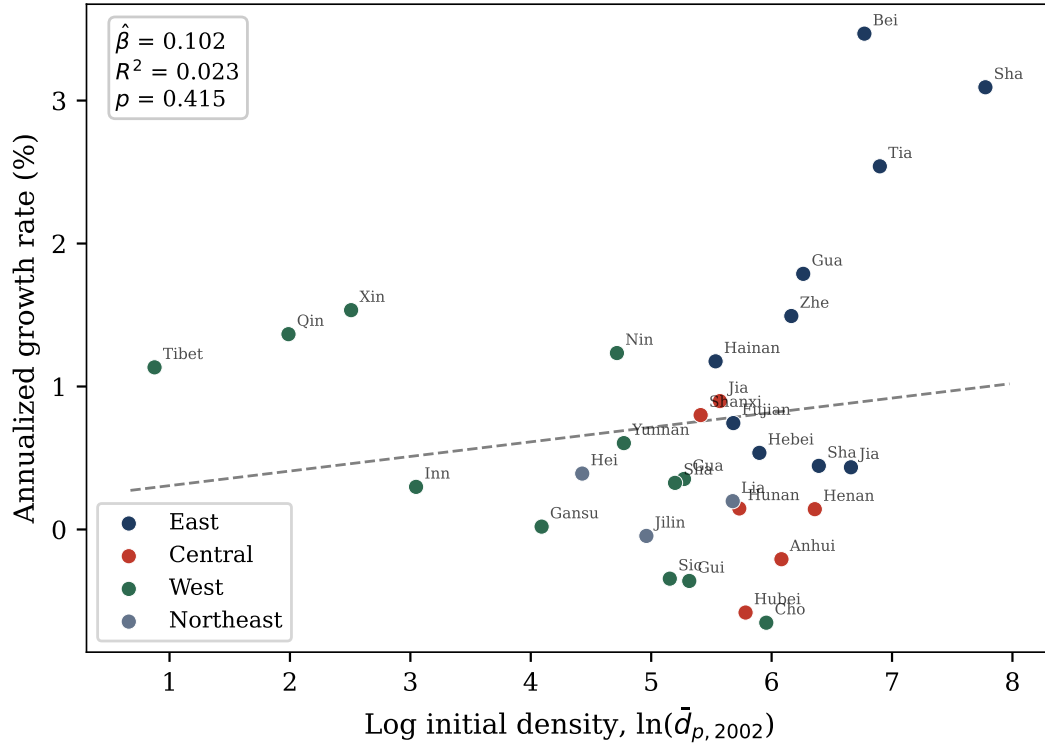


Figure 5: β -convergence: annualized density growth rate versus log initial density, by province and region, 2002–2020.

σ -convergence. The coefficient of variation of provincial mean densities increased monotonically from 1.176 in 2002 to 1.540 in 2020, indicating that the gap between dense and sparse provinces widened in every year of the sample rather than narrowing (the annual CV series is plotted in Appendix Figure 6). The evidence rejects σ -convergence over the full sample period.

Table 2: Convergence diagnostics, 2002–2020

Indicator	Value	Interpretation
$\hat{\beta}$ (OLS slope)	0.1021	Divergence
Standard error	0.1234	
R^2	0.0231	
p -value	0.4149	Not significant at 5%
CV (2002)	1.1756	
CV (2020)	1.5401	
CV change	0.3645	

Notes: $\hat{\beta}$ is the OLS slope from regressing annualized provincial density growth on log initial density ($\ln \bar{d}_{p,2002}$). CV is the coefficient of variation of provincial mean densities. Excludes Hong Kong, Macao, and Taiwan.

Table 3 provides province-level detail. The fastest-growing provinces by density include several coastal and south-central provinces, while several northeastern provinces experienced slower growth. This pattern is broadly consistent with the centroid evidence and with the uneven pull of market access across Chinese regions.

5 Discussion

I interpret the four findings using spatial economics, consider alternative explanations, and note limitations.

5.1 Agglomeration and core-periphery dynamics

Rising concentration and the outward expansion of dense cores are consistent with standard agglomeration forces. In the core-periphery model (Krugman, 1991), increasing returns, transport costs, and market size interact so that as more workers concentrate in a region, the local market expands, firms enjoy scale economies, and the region becomes more attractive. A rising Gini, growing top-decile concentration, and an expanding geographic footprint of high-density areas match this prediction.

Desmet and Rossi-Hansberg (2014) show that spatial reallocation can be a long-run equilibrium process driven by innovation and migration. The 18-year centroid trajectory in

Table 3: Provincial mean density and annualized growth, 2002–2020

Province	Region	Mean density 2002 (pers./km ²)	Mean density 2020 (pers./km ²)	Ann. growth rate (%)
Beijing	East	871.6	1610.1	3.47
Fujian	East	293.5	335.4	0.74
Guangdong	East	524.5	721.6	1.79
Hainan	East	253.3	312.7	1.18
Hebei	East	364.4	401.2	0.54
Jiangsu	East	779.5	842.8	0.43
Shandong	East	597.0	646.6	0.44
Shanghai	East	2381.1	4120.0	3.09
Tianjin	East	990.9	1556.4	2.54
Zhejiang	East	475.2	620.5	1.49
Anhui	Central	437.7	421.6	−0.21
Henan	Central	577.5	592.5	0.14
Hubei	Central	324.9	292.6	−0.58
Hunan	Central	308.4	316.7	0.15
Jiangxi	Central	262.6	308.5	0.90
Shanxi	Central	223.8	258.3	0.80
Chongqing	West	386.0	343.1	−0.65
Gansu	West	59.7	60.0	0.02
Guangxi	West	194.9	207.7	0.35
Guizhou	West	203.6	190.8	−0.36
Inner Mongolia	West	21.1	22.2	0.30
Ningxia	West	111.7	139.2	1.23
Qinghai	West	7.3	9.3	1.37
Shaanxi	West	180.9	191.8	0.33
Sichuan	West	173.2	162.8	−0.34
Tibet	West	2.4	2.9	1.13
Xinjiang	West	12.3	16.1	1.53
Yunnan	West	118.3	131.8	0.60
Heilongjiang	Northeast	83.7	89.8	0.39
Jilin	Northeast	142.6	141.4	−0.05
Liaoning	Northeast	292.1	302.7	0.20

Notes: Mean density is the unweighted average of pixel-level density within each province. Annualized growth rate is computed as $g_p = (d_{p,2020}/d_{p,2002})^{1/18} - 1$. Excludes Hong Kong, Macao, and Taiwan.

Section 4 is consistent with such gradual adjustment rather than a one-time shock.

Agglomeration also generates congestion costs—rising housing prices and commuting burdens—that limit the growth of the largest cities (Combes et al., 2019; Au and Henderson, 2006). That concentration rose but did not accelerate is consistent with agglomeration pull coexisting with increasing congestion in the densest cores.

5.2 Market access and directional reallocation

The southeastern centroid drift may reflect, among other factors, differential changes in market access along China’s eastern seaboard and in the Pearl River Delta and Yangtze River Delta. Redding and Sturm (2008) show that market access is a primary determinant of the spatial distribution of economic activity; when access improves differentially, population and production relocate toward better-connected areas. Zheng and Kahn (2013) document that high-speed rail in China facilitated market integration and reduced the cost of megacity growth, which may have reinforced southeastern concentration.

That the centroid moved in a consistent direction over 18 years, rather than erratically, is more consistent with broader structural forces—market access, employment growth, public-service provision—than with purely idiosyncratic local shocks.

5.3 Why convergence was absent

In the growth-convergence literature (Barro and Sala-i Martin, 1992), a negative cross-sectional relationship between initial levels and subsequent growth indicates that poorer (or less dense) regions are catching up. This relationship is absent here, suggesting that the mechanisms producing income convergence in some settings do not apply directly to the spatial distribution of population in China.

Three factors may explain the absence of convergence. First, the hukou system and related institutional frictions (Chan and Zhang, 1999) restrict labor mobility, slowing the reallocation that would otherwise reduce spatial dispersion. Second, as Kanbur and Zhang (2005) and Fan and Sun (2008) document, the relative importance of coastal-interior, urban-rural, and interprovincial inequality shifted over time, so the aggregate convergence pattern depends on which forces dominate in a given period. Third, if agglomeration forces are strong relative to diminishing returns, dispersion can persist or widen rather than narrow (Fujita et al., 1999). The evidence here is more suggestive of agglomeration forces outweighing catch-up dynamics than of broad-based spatial equalization.

5.4 Limitations

First, population density is not income, productivity, or welfare. Rising concentration does not by itself imply that residents of dense areas are better off or that residents of sparse areas are worse off. Second, the WorldPop rasters are model-based estimates with inherent uncertainty (Stevens et al., 2015); they measure broad spatial patterns, not exact local counts. Third, the analysis is descriptive. The patterns are consistent with agglomeration forces, market access, and institutional frictions, but the data do not identify any of these channels causally. Separating them would require exogenous variation in transport infrastructure, migration policy, or market integration—a different exercise from the one undertaken here.

Fourth, the downsampled 5 km grid trades spatial resolution for computational tractability. While this is appropriate for national and provincial analysis, it smooths over within-city heterogeneity. Finally, the convergence analysis uses a simple two-period framework. More sophisticated panel approaches could reveal different patterns within subperiods.

6 Conclusion

China’s population became more spatially concentrated between 2002 and 2020. Density gains accrued disproportionately to areas that were already dense, the geographic footprint of high-density cores expanded outward from established clusters, and the national population centroid drifted steadily toward the southeast. At the province level, neither β - nor σ -convergence appears in the data: initially sparse provinces did not grow faster, and cross-provincial dispersion widened.

These patterns bear on the tension between agglomeration theory and convergence theory as applied to population. Standard agglomeration models predict that increasing returns and market-access advantages concentrate activity in a few locations; standard convergence models predict that diminishing returns eventually equalize levels across regions. The coexistence of rising pixel-level concentration and rising provincial dispersion suggests that, over this period, agglomeration forces dominated any tendency toward spatial catch-up. If this reading is correct, it suggests that policies aimed at equalizing population across provinces may work against persistent agglomeration forces rather than simply accelerating a natural convergence process.

The descriptive evidence leaves open the question of which specific mechanisms—hukou reform, transport investment, housing-supply constraints, or differential productivity growth—account for the patterns documented here. Disentangling these channels requires exogenous variation that a descriptive exercise cannot supply, but the spatial facts reported in this

paper provide a benchmark against which structural estimates of these forces can be compared. A companion website (<https://cpde.streamlit.app>) makes the maps and annual indicators available for interactive inspection.

Data availability

The WorldPop UN-adjusted population density rasters are publicly available at <https://www.worldpop.org>. The province-boundary GeoJSON and replication code for all figures, tables, and indicators reported in this paper are available at <https://pop-density-spatial.streamlit.app/>.

References

- Ahlfeldt, G. M., Redding, S. J., Sturm, D. M., and Wolf, N. (2015). The economics of density: Evidence from the Berlin wall. *Econometrica*, 83(6):2127–2189.
- Allen, T. and Arkolakis, C. (2014). Trade and the topography of the spatial economy. *Quarterly Journal of Economics*, 129(3):1085–1140.
- Au, C.-C. and Henderson, J. V. (2006). Are Chinese cities too small? *Review of Economic Studies*, 73(3):549–576.
- Barro, R. J. and Sala-i Martin, X. (1992). Convergence. *Journal of Political Economy*, 100(2):223–251.
- Baum-Snow, N., Brandt, L., Henderson, J. V., Turner, M. A., and Zhang, Q. (2017). Roads, railroads, and decentralization of Chinese cities. *Review of Economics and Statistics*, 99(3):435–448.
- Brandt, L., Ma, D., and Rawski, T. G. (2014). From divergence to convergence: Reevaluating the history behind China’s economic boom. *Journal of Economic Literature*, 52(1):45–123.
- Bryan, G. and Morten, M. (2019). The aggregate productivity effects of internal migration: Evidence from Indonesia. *Journal of Political Economy*, 127(5):2229–2268.
- Caliendo, L., Parro, F., Rossi-Hansberg, E., and Sarte, P.-D. (2018). The impact of regional and sectoral productivity changes on the U.S. economy. *Review of Economic Studies*, 85(4):2042–2096.

- Chan, K. W. and Zhang, L. (1999). The hukou system and rural-urban migration in China: Processes and changes. *The China Quarterly*, 160:818–855.
- Chauvin, J. P., Glaeser, E., Ma, Y., and Tobio, K. (2017). What is different about urbanization in rich and poor countries? Cities in Brazil, China, India and the United States. *Journal of Urban Economics*, 98:17–49.
- Combes, P.-P., Duranton, G., and Gobillon, L. (2019). The costs of agglomeration: House and land prices in French cities. *Review of Economic Studies*, 86(4):1556–1589.
- Desmet, K. and Rossi-Hansberg, E. (2014). Spatial development. *American Economic Review*, 104(4):1211–1243.
- Dingel, J. I., Miscio, A., and Davis, D. R. (2021). Cities, lights, and skills in developing economies. *Journal of Urban Economics*, 125:103174.
- Donaldson, D. and Hornbeck, R. (2016). Railroads and American economic growth: A “market access” approach. *Quarterly Journal of Economics*, 131(2):799–858.
- Faber, B. (2014). Trade integration, market size, and industrialization: Evidence from China’s national trunk highway system. *Review of Economic Studies*, 81(3):1046–1070.
- Fan, C. C. and Sun, M. (2008). Regional inequality in China, 1978–2006. *Eurasian Geography and Economics*, 49(1):1–18.
- Fujita, M., Krugman, P., and Venables, A. J. (1999). *The Spatial Economy: Cities, Regions, and International Trade*. MIT Press.
- Henderson, J. V., Storeygard, A., and Weil, D. N. (2012). Measuring economic growth from outer space. *American Economic Review*, 102(2):994–1028.
- Hsieh, C.-T. and Moretti, E. (2019). Housing constraints and spatial misallocation. *American Economic Journal: Macroeconomics*, 11(2):1–39.
- Imbert, C., Seror, M., Zhang, Y., and Zylberberg, Y. (2022). Migrants and firms: Evidence from China. *American Economic Review*, 112(6):1885–1914.
- Kanbur, R. and Zhang, X. (2005). Fifty years of regional inequality in China: A journey through central planning, reform, and openness. *Review of Development Economics*, 9(1):87–106.
- Kennan, J. and Walker, J. R. (2011). The effect of expected income on individual migration decisions. *Econometrica*, 79(1):211–251.

- Kline, P. and Moretti, E. (2014). Local economic development, agglomeration economies, and the big push: 100 years of evidence from the Tennessee Valley Authority. *Quarterly Journal of Economics*, 129(1):275–331.
- Krugman, P. (1991). Increasing returns and economic geography. *Journal of Political Economy*, 99(3):483–499.
- Lessmann, C. and Seidel, A. (2017). Regional inequality, convergence, and its determinants – a view from outer space. *European Economic Review*, 92:110–132.
- Michaels, G., Rauch, F., and Redding, S. J. (2012). Urbanization and structural transformation. *Quarterly Journal of Economics*, 127(2):535–586.
- Poncet, S. (2005). A fragmented China: Measure and determinants of chinese domestic market disintegration. *Review of International Economics*, 13(3):409–430.
- Redding, S. J. and Rossi-Hansberg, E. (2017). Quantitative spatial economics. *Annual Review of Economics*, 9:21–58.
- Redding, S. J. and Sturm, D. M. (2008). The costs of remoteness: Evidence from German division and reunification. *American Economic Review*, 98(5):1766–1797.
- Stevens, F. R., Gaughan, A. E., Linard, C., and Tatem, A. J. (2015). Disaggregating census data for population mapping using random forests with remotely-sensed and ancillary data. *PLoS ONE*, 10(2):e0107042.
- Tombe, T. and Zhu, X. (2019). Trade, migration, and productivity: A quantitative analysis of China. *American Economic Review*, 109(5):1843–1872.
- WorldPop and CIESIN (2018). Global high resolution population denominators project.
- Zheng, S. and Kahn, M. E. (2013). China’s bullet trains facilitate market integration and mitigate the cost of megacity growth. *Proceedings of the National Academy of Sciences*, 110(14):E1248–E1253.

A Additional Figures

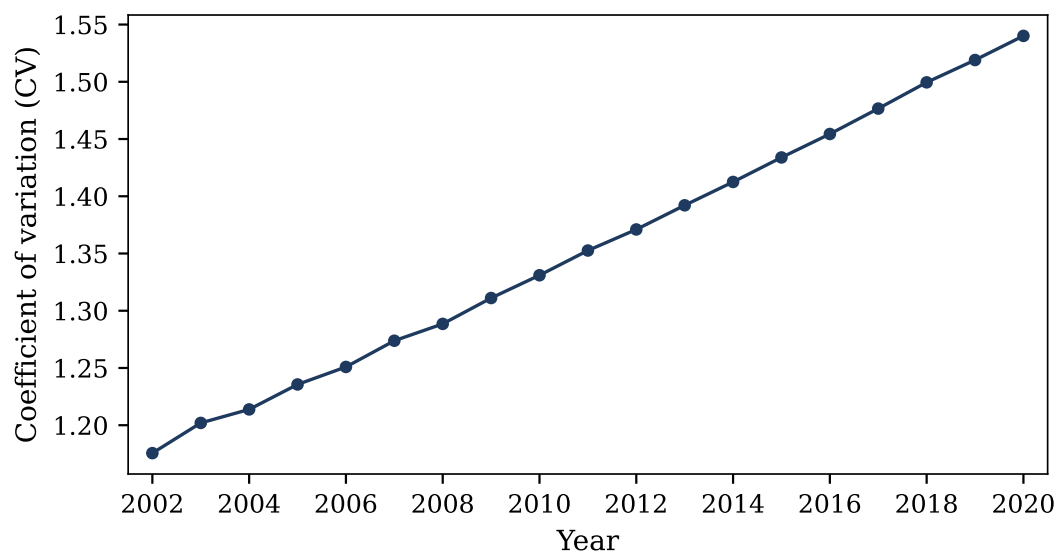


Figure 6: Cross-provincial dispersion in mean density (CV), mainland China, 2002–2020.